AFDB · DATA INNOVATION FOR AFRICA (DI4A)

# Get a NASA Earthdata token for night-lights data

*Access NASA Black Marble (VNP46) data in six steps*



AFRICAN DEVELOPMENT BANK
Data Innovation
for Africa (DI4A)

**African Development Bank (AfDB)**
Data Innovation for Africa (DI4A)

# One token unlocks NASA Black Marble

**Earthdata account**
Free NASA login

→

**Bearer token**
Your access key, valid 60 days

→

**LAADS DAAC**
NASA archive hosting VNP46

→

**blackmarblepy**
Python / R: download and clip

→

**NTL series**
Monthly national indicators

## Black Marble products (VIIRS, ~500 m)

| Product | Content |
|---|---|
| VNP46A1 | Daily, at-sensor radiance |
| VNP46A2 | Daily, moonlight-corrected, gap-filled |
| VNP46A3 | Monthly composite — used in the analysis |
| VNP46A4 | Annual composite |

### Where you will use it

AfDB · DI4A night-lights analysis: download and clip NASA Black Marble monthly data, then build national night-lights indicators.

# From a NASA account to a working token

01

**Register**

urs.earthdata.nasa.gov, with a complete profile

02

**Activate**

Click the link in the confirmation email

03

**Accept EULAs**

MERIS and Sentinel-3, from your profile

04

**Authorize LAADS**

Approve the application on ladsweb

05

**Generate token**

Profile › Generate Token › Show Token

06

**Test and use**

curl test, then blackmarblepy / blackmarbler

Most token failures come from steps 01, 03 and 04: an incomplete profile, missing EULAs or a LAADS app not authorized.

# Create your Earthdata Login account

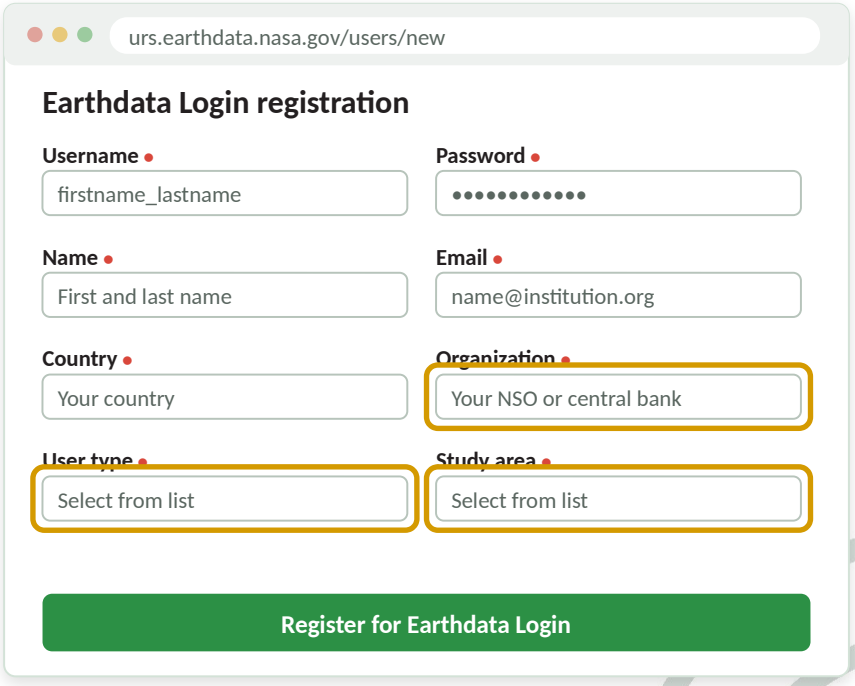
1. **Open urs.earthdata.nasa.gov**
   Click Register (or on ladsweb: Login › Earthdata Login).

2. **Choose a username and password**
   You will reuse them to log in to NASA sites.

3. **Fill the required fields**
   They are marked with red dots: name, email, country.

4. **Do not skip the highlighted fields**
   Organization, user type and study area: the token fails without them.

5. **Click Register for Earthdata Login**
   Then go to your mailbox.



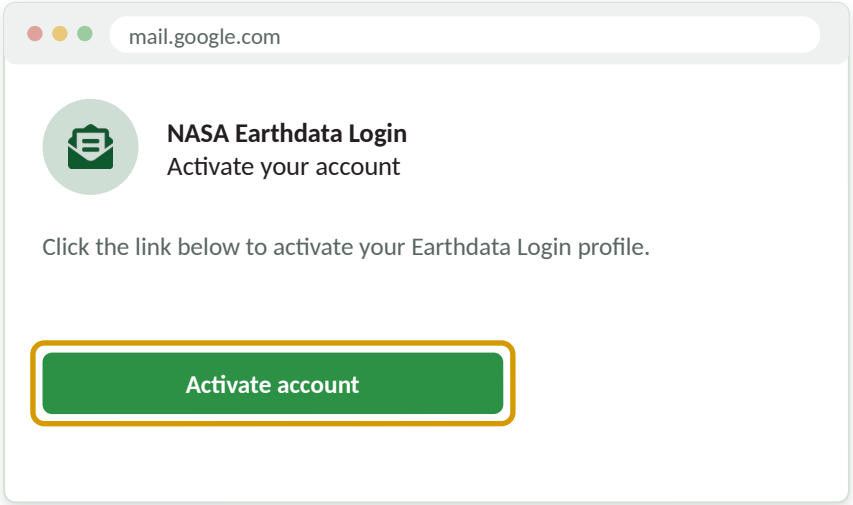

*Schematic illustration — the live page may look slightly different.*

# Activate your account by email

1. **Open your mailbox**
   Look for the message from Earthdata Login (check Spam).

2. **Click the activation link**
   Your profile is activated.

3. **Log in**
   At urs.earthdata.nasa.gov with your username and password.




No email after 10 minutes? Check Spam or Promotions, then ask the support team.

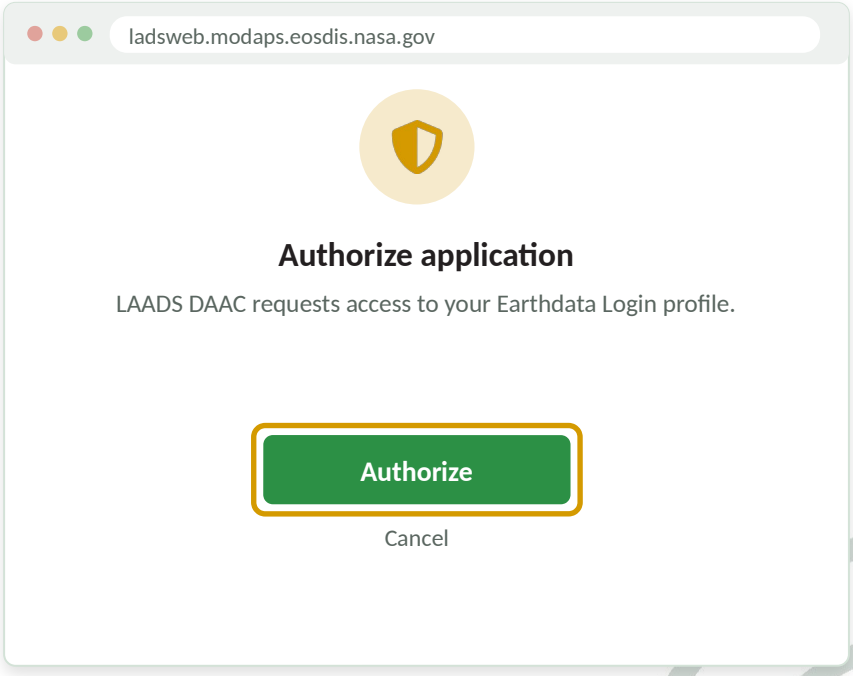
*Schematic illustration — the live page may look slightly different.*

# Authorize the LAADS DAAC application

1. **Log in to LAADS DAAC**
   ladsweb.modaps.eosdis.nasa.gov › Login › Earthdata Login.

2. **Open a Black Marble product page**
   For example VNP46A4 or VNP46A3.

3. **Click the green Authorize button**
   It lets LAADS use your Earthdata profile (Sentinel-3 and MERIS).




*Schematic illustration — the live page may look slightly different.*

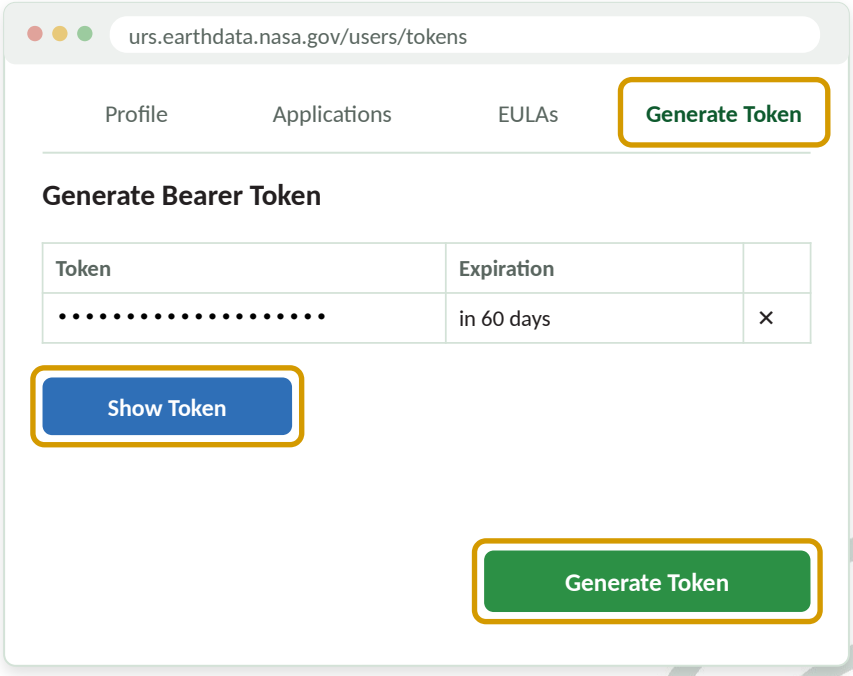
# Generate and copy your token

1. **Log in to your Earthdata profile**
   urs.earthdata.nasa.gov/profile.

2. **Click Generate Token**
   In the top menu of the profile.

3. **Click the green Generate Token button**
   At the bottom of the page, if no token is listed yet.

4. **Click Show Token, then copy it**
   Paste it straight into a password manager or an environment variable.




*Schematic illustration — the live page may look slightly different.*

# Three facts about your token

**60 days**

Each user token is valid for 60 days, then you generate a new one.

**2 tokens max**

At most two valid tokens per user at any time. Delete one to create another.

**1 header**

Send it with every request as: Authorization: Bearer <token>.

Treat the token like a password: never paste it into a shared notebook, a slide or a GitHub repository. Use an environment variable.

*Sources: Earthdata Login documentation, "User tokens"; LAADS DAAC, "Download files using EDL tokens".*

# Store the token, test it, download Black Marble

### STORE IT IN AN ENVIRONMENT VARIABLE

```
# macOS / Linux terminal
export BLACKMARBLE_TOKEN="eyJ0eXAi..."
# Windows PowerShell
$env:BLACKMARBLE_TOKEN="eyJ0eXAi..."
# Google Colab / Jupyter
import os
os.environ["BLACKMARBLE_TOKEN"] = "eyJ..."
```

### TEST IT: DOWNLOAD THE LAADS README

```
# Should return a text file, not an HTML login page
curl -L -b session \
  -H "Authorization: Bearer $BLACKMARBLE_TOKEN" \
  -o README.txt \
  https://ladsweb.modaps.eosdis.nasa.gov/archive/README
```

### USE IT: MONTHLY NIGHT LIGHTS (PYTHON)

```
# pip install blackmarblepy
import geopandas as gpd
import pandas as pd
from blackmarble import BlackMarble, Product

gdf = gpd.read_file("my_country.geojson")
bm = BlackMarble()  # reads BLACKMARBLE_TOKEN

ntl = bm.raster(
    gdf,
    product_id=Product.VNP46A3,  # monthly
    date_range=pd.date_range(
        "2025-01-01", "2025-12-01", freq="MS"),
)
```

*R users: blackmarbler::bm_raster(roi_sf, product_id = "VNP46A3", date = "2025-01-01", bearer = token).*

# If the token does not work

### Access denied (401)

Check the profile fields (organization, user type, study area), the MERIS and Sentinel-3 EULAs and the LAADS authorization.

### Token expired

Tokens last 60 days. Generate a new one and update BLACKMARBLE_TOKEN.

### Cannot create a new token

You already have two valid tokens. Delete one with ✕, then generate again.

### curl returns an HTML page

Keep -L and -b session, use -H (not -h) and a space after “Bearer”.

# Sources and references

**Earthdata Login — User tokens**
urs.earthdata.nasa.gov/documentation/for_users/user_token

**LAADS DAAC — Download files using EDL tokens**
ladsweb.modaps.eosdis.nasa.gov/learn/download-files-using-edl-tokens/

**World Bank — blackmarbler (R): NASA bearer token steps**
worldbank.github.io/blackmarbler/

**blackmarblepy (Python package)**
pypi.org/project/blackmarblepy/

*Procedure checked in October 2026; NASA interfaces may change.*

YOU ARE READY

# Your token is ready for Black Marble

*Keep it at hand for your night-lights analysis (AfDB · DI4A).*

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